

Mortadha Ben Younes

Industrial engineering student

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SUMMARY OF QUALIFICATIONS

- Highly motivated Industrial Engineering student with a dual qualification in energy systems and economics, and a strong academic and research background in supply chain optimization and data-driven decision-making.
- Proven background in energy optimization and techno-economic solutions, from renewable energy system sizing to feasibility and simulation-based studies across the energy value chain.
- Strong analytical skills in mathematical modeling and system simulation, with a focus on rigor across research and industrial projects.
- Excellent organizational and time-management skills, demonstrated through various academic and professional projects.

EDUCATION

International Master Program on Renewable Energy Systems for Africa – TEchnology And Management (dual enrollement) 2026

National Engineering School of Tunis

Relevant Coursework : Carbon Market , Circular Economy , Blockchain applied to energy, Optimization, modeling and prospective evaluation for energy systems , Project management

National Diploma in Engineering -Industrial Engineering 09/2024 – Present | Manar, Tunis

National engineering School of Tunis 

Relevant Coursework : Supply Chain Management, Operations Research, Stochastic Processes, Operations Management, Data Simulation , ERP integrated course , Data Mining

Preparatory Cycle Diploma for Engineering Studies (Physics-Chemistry) 09/2021 – 06/2024 | Monastir

Preparatory Institute for Engineering Studies of Monastir (IPEIM) 

Rank : 151/1230

High School Diploma in Experimental Sciences 09/2019 – 06/2020 | Monastir

Pioneer High School of Monastir

ACADEMIC PROJECTS

Green Port Decarbonization Hub & E-Methanol Production, Academic Project 2026

- Designed a green methanol production hub to fuel Mediterranean cruise ships
- Dimensioned a green hydrogen system using solar power, PEM electrolysis, and CO² hydrogenation
- Planned direct port-side bunkering logistics to eliminate transport costs and meet EU emissions standards.

Techno-Economic Feasibility Study of Advanced Energy Storage System, Academic Project 2024

- Evaluated diverse storage technologies (Li-ion, thermal, and hydrogen) based on efficiency, lifecycle, and levelized cost of storage (LCOS).

Final-Year Research Project (PFA2): Development of a Simulation Model for the Joint Optimization of Quality and Maintenance in Industrial Systems 2026

- Developed a discrete-event simulation model using ARENA software to analyze the stochastic behavior of industrial production lines.

- Integrated quality-maintenance coupling by implementing an X-bar control chart synchronized with machine degradation states.
- Optimized maintenance strategies (preventive vs. corrective) to minimize the Total Expected Cost (TEC) and maximize system availability

Final-Year Research Project (PFA1): Strategic Study – Innovation and Challenges in the Banking Sector

2025

- Analyzed the impact of digital transformation and emerging financial models (Fintech) on traditional banking.
- Developed a comprehensive understanding of risk management and investment trends in evolving markets

 PROFESSIONAL EXPERIENCES

Tunisian National Electricity and Gas Company (STEG)

07/2026 – Present | Sousse

Engineer Internship

- System Sizing & Modeling: Sizing and modeling utility-scale Photovoltaic (PV) systems integrated with Siemens Silyzer electrolyzers for green H² production.
- Techno-Economic Optimization: Conducting energy yield simulations and dynamic load balancing to maximize green hydrogen output and system efficiency.
- Grid Integration Analysis: Assessing technical feasibility, multi-vector energy flows, and capacity limits for coupling electrolysis with the national power grid.

Shadowing Internship – Impression & Teinture du Sahel (ITS)

June 2025

Maintenance Departement

- Used a Computerized Maintenance Management System (CMMS) for maintenance operations.
- Collaborated with the company’s maintenance team on equipment servicing and upkeep

Hotel AMIR PALACE , *Operations & Quality Controller*

07/2023 – 08/2023 | Monastir

- Conducted daily and monthly physical inventories to reconcile actual stock with digital records in the "Économat" department.

 TOOLS & CERTIFICATIONS

Tools: ARENA , Excel , Lingo , PVGIS , Python , SAP , CAO

Certifications: Renewable Power to X Basic Training (International PtX Hub) , Lean Management , Injaz Company , Negotiation Technics

 EXTRACURRICULAR ACTIVITIES

Industrial Engineering Club – ENIT, Senior Member

09/2024 – Present

- Attended conferences on ethics and professional conduct in engineering.
- Organized and participated in industrial site visits to analyze production lines and supply chains

ENIT Junior Enterprise, Active Member

09/2024 – 06/2025

- Business development and prospection: identified and negotiated industrial partnerships and conducted market research.
- Implemented internal process standardization in compliance with JET standards to ensure quality deliverables

 LANGUAGES

English TOEIC B2	French DEL F B2	Espagnol Basic notions	Arabic Native speaker
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